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Body Mass Index, Physical Activity, and Epigenetic Aging: A Cross-Population Study

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Abstract Background: Observational studies have suggested that body mass index (BMI) is associated with accelerated epigenetic aging; however, the age at which this association begins and the duration of its effects have not been fully established. This study evaluated the effects of BMI and physical activity on epigenetic aging from early life to adulthood. **Methods:** We integrated multi-source data from populations of different racial backgrounds, including genome-wide association study (GWAS) data, the National Health and Nutrition Examination Survey (NHANES), clinical visceral adipose tissue, and blood samples from obese and physically active populations. Specifically, Mendelian randomization (MR) analysis was used to infer the causal relationship between early-life BMI and epigenetic age acceleration and to identify potential mediators. NHANES data were then analyzed to validate this association at the population level. Finally, biomarkers in blood and adipose tissue were examined to assess whether physical activity is associated with reduced aging-related signatures. **Results:** The Mendelian randomization analyses indicated that high early-life BMI was causally associated with accelerated epigenetic aging across multiple epigenetic clocks, with β ranging from 0.231 to 0.506; similar associations were also observed for childhood obesity, with β ranging from 0.139 to 0.231. Physical activity was identified as an important mediator that attenuated epigenetic aging acceleration, with mediation effects accounting for 10.70% to 16.22%. Analysis of the NHANES dataset further confirmed that epigenetic age acceleration increases progressively with increasing BMI: [$25 \leq$ BMI < 30: β=0.83, 95% CI (0.66, 1.00); $30 \leq$ BMI < 35: β=1.73, 95% CI (1.54, 1.93); $35 \leq$ BMI < 40: β=2.93, 95% CI (2.67, 3.20); BMI ≥ 40: β=4.64, 95% CI (4.30, 4.99)]. Subgroup analysis showed that, at the same BMI level, female exhibited more severe PhenoAA acceleration than male. Validation using multi-tissue samples showed that obesity accelerates DNA damage and significantly upregulates genes involved in cell cycle regulation. Physical activity significantly inhibited the upregulation of aging-related genes and reduced oxidative stress. **Conclusion:** BMI is strongly associated with accelerated epigenetic aging. This association is evident in children with obesity, becomes stronger with higher BMI, and is also observed in adults, whereas physical activity may partially attenuate this process.

Keywords: Epigenetics; Aging clock; Physical activity; Body mass index

From 1990 to 2021, the global prevalence of overweight and obesity has consistently increased, with all countries showing an upward trend. By 2021, approximately 1 billion male and 1.11 billion female worldwide were overweight or obese[1]. Obesity increases the risk of chronic diseases and accelerated aging[2]. With population aging becoming a growing global concern, the prevention and delay of aging have attracted widespread attention in the scientific community. In this context, the distinction between chronological and biological age is particularly important, as individual aging processes often deviate from calendar time. Chronological age refers to the actual time a person has lived, usually measured in years. In contrast, biological age reflects an individual's physiological and functional status, as evaluated using multiple indicators, including cardiopulmonary function, hormone levels, inflammatory markers, and epigenetic aging clocks. People of different genders and ethnicities may have the same chronological age, but their biological age may differ significantly, often related to environmental factors, daily behaviors, and endogenous or exogenous stresses[3]. Emerging evidence suggests that biological age is strongly associated with the risk of multiple chronic diseases, including cancer, metabolic disorders, and cardiovascular disease. Therefore, assessing biological age may provide a clinically meaningful framework for disease prevention and aging surveillance, while also helping to clarify how modifiable factors, such as obesity and physical activity, may accelerate or delay the aging process[4].

Aging is a complex process involving changes in gene expression, protein synthesis, hormone secretion, and cellular function[5]. Within this complex aging process, epigenetic modifications play a crucial role by regulating gene activity and expression, thereby influencing cell and tissue function. Epigenetic "age estimators" use specific CpG sites (also known as epigenetic clocks) and mathematical algorithms to estimate DNA methylation age. Due to its applicability across various tissues, high precision, and strong correlation, this method has been recognized as one of the most promising tools for predicting biological age[6][7]. Numerous studies have shown that obesity is significantly associated with cardiovascular and metabolic diseases, both in childhood and adulthood. This association may operate through epigenetic mechanisms. Research has demonstrated that the liver and visceral fat tissue significantly accelerate epigenetic aging in obese states[8][9][10]. More importantly, this accelerated aging process has been directly linked to an increased risk of coronary artery disease and type 2 diabetes, with a particularly pronounced effect in obese populations[11]. Conversely, regular physical activity promotes the secretion of exerkines from multiple organs, thereby modulating DNA methylation patterns and contributing to a delay in epigenetic aging[12][13]. This may partly explain why individuals who engage in regular physical activity often exhibit a lower biological age than age-matched individuals with obesity[14][15]. Taken together, these findings highlight the potential role of obesity and physical activity as key modifiable determinants of biological aging and underscore the importance of weight management for both delaying aging and preventing related chronic diseases.

Early-life body mass index (BMI) may better capture the obesity-related health burden[16]. Among adults with high levels of physical activity, an elevated BMI can partly reflect greater lean and bone mass rather than excess adiposity; by contrast, early-life BMI is less likely to be substantially influenced by these factors[17]. Accordingly, using early-life BMI as the exposure may provide a clearer assessment of the relationship between adiposity-related BMI variation and epigenetic aging, thereby informing intervention strategies to decelerate aging and mitigate obesity-related health risks. To further advance this field, we aimed to determine whether early-life BMI is causally associated with accelerated epigenetic aging, examine associations across BMI levels, and evaluate whether physical activity is associated with slower epigenetic aging.

2. Methods

2.1 Overall Study Design

We adopted an integrated study design. In the first phase, Mendelian randomization (MR) was employed to establish a causal relationship between early-life BMI and accelerated epigenetic aging, followed by mediation analysis to identify mediating factors that potentially decelerate this process. In the second phase, the NHANES database was utilized to further validate the association between BMI and epigenetic aging. Finally, blood samples from exercising cohorts and visceral fat samples were analyzed to determine the molecular-level effects of physical activity on slowing aging (Figure 1). By combining public databases with clinical samples, we aimed to demonstrate the multi-dimensional impact of obesity and physical activity on epigenetic aging.

2.2 Mendelian Randomization Analysis

2.2.1 Basic Concepts of MR Analysis

Since genetic variants are randomly allocated at conception and are generally not influenced by external environmental factors, MR analysis is less susceptible to confounding and reverse causation. Therefore, in this study, MR analysis was used to investigate the relationship between early-life BMI and epigenetic age acceleration using single-nucleotide polymorphisms (SNPs) associated with the exposure of interest. To obtain valid causal estimates, the genetic variants used as instrumental variables (IVs) in MR analysis must satisfy three core assumptions (Figure 1)[18]. First, the selected SNPs must be strongly associated with early-life BMI. Second, these genetic variants must not be confounded by other factors. Third, their effects on epigenetic age acceleration must operate exclusively through early-life BMI.

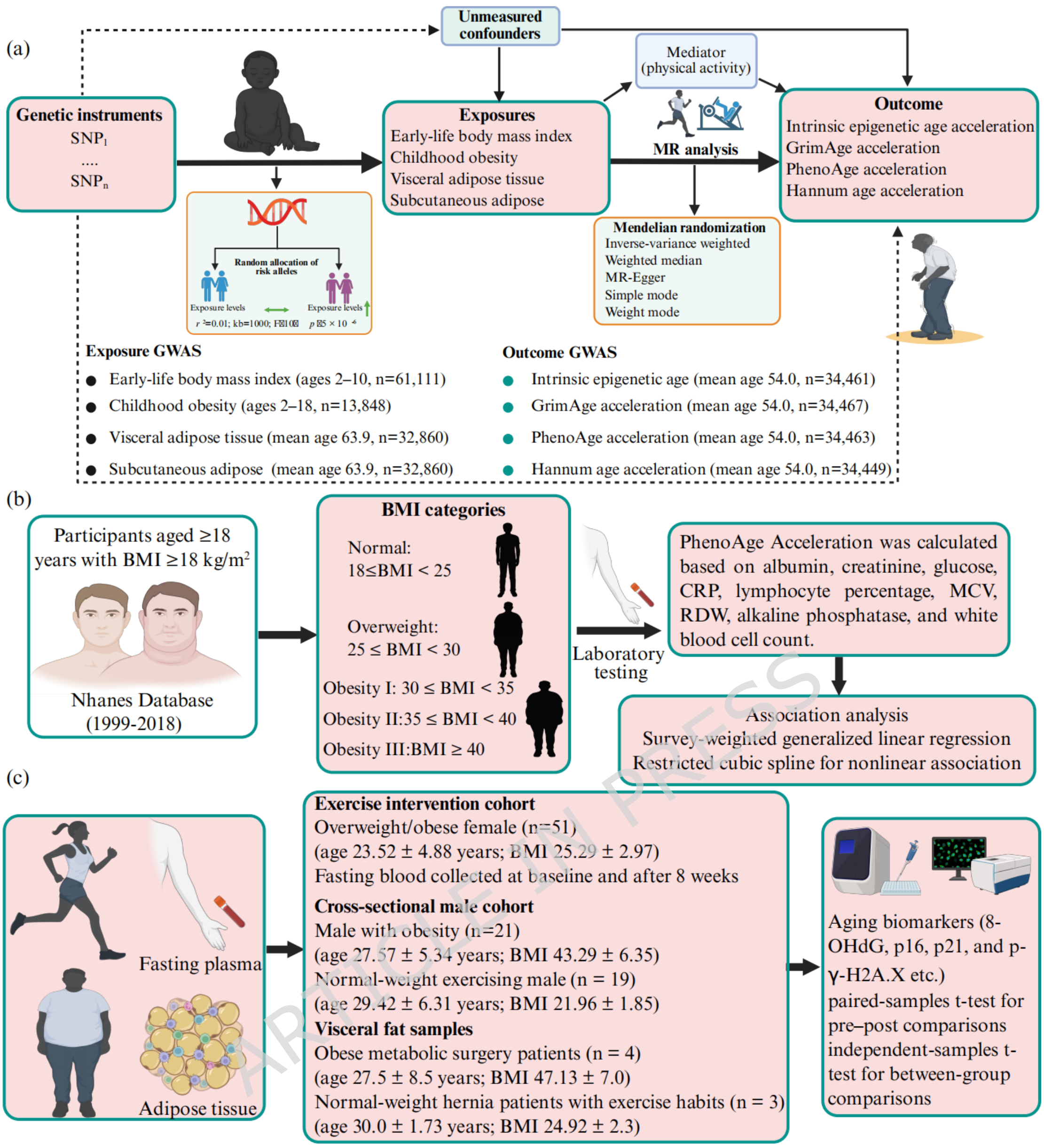

Figure 1.Study design. (a) Schematic overview of the Mendelian randomization (MR) study design. Genetic variants (SNPs) associated with the exposures were used as instrumental variables to estimate the causal effects of early-life body mass index, childhood obesity, visceral adiposity, and subcutaneous adiposity on four epigenetic age acceleration measures. (b) Design of the NHANES population-based analysis. Participants aged ⩾18 years with BMI ⩾18 were selected from the NHANES database (1999-2018). Individuals were stratified by BMI categories to evaluate the association between BMI and epigenetic age. (c) Clinical sample collection and molecular assays. Adipose tissue and fasting plasma samples were collected for molecular measurements of aging-related biomarkers. The representative human figures and sample illustrations in the figure were created with BioRender.com.

2.2.2 Data Sources

We used a two-sample MR analysis to investigate the causal effect of early-life

BMI on epigenetic age acceleration. Early-childhood BMI (ages 2–10)[19] and childhood obesity (ages 2–18)[20], visceral adipose tissue, and abdominal subcutaneous adipose tissue were treated as exposures[21]. The outcomes were four epigenetic aging clocks（average age 54）, including Intrinsic Epigenetic Age Acceleration (IEAA), GrimAge Acceleration (GrimAA), Hannum Age Acceleration (HannumAA), and PhenoAge Acceleration (PhenoAA)[22]. In addition, physical activity was evaluated as a potential mediator[23]. Summary statistics for all traits were obtained from large-scale genome-wide association studies (GWAS) (Table 2, Supplementary File 1).

2.2.3 SNP Selection for MR Analysis

When selecting instrumental variables, we set the genome-wide significance threshold at $p < 5 \times 10^{-6}$ to obtain a larger number of SNPs. After excluding SNPs in linkage disequilibrium using an r^2 threshold of < 0.001 within a 10,000 kb window, the remaining independent SNPs were retained as instrumental variables for early-childhood BMI, childhood obesity, visceral adipose tissue, and abdominal subcutaneous adipose tissue[19–21]. Their associations with the four epigenetic aging clocks were then extracted from the corresponding GWAS summary statistics[22]. Finally, we calculated the F-statistic for each SNP to assess instrument strength and excluded weak instruments with F-statistics < 10 (SNPs are shown in Supplementary Material File 2).

2.2.4 Mediation Analysis

To further explore whether physical activity mediated the association between early-life adiposity and epigenetic aging, we performed a two-step MR mediation analysis[24]. This approach estimates the causal effect of the exposure on the mediator and the causal effect of the mediator on the outcome, thereby allowing the indirect effect through the mediator to be quantified.

2.3.1 Data Source and Study Population

The data used in this study were collected from the National Health and Nutrition Examination Survey (NHANES) conducted between 1999 and 2018. The data, sourced from the National Center for Health Statistics, Centers for Disease Control and Prevention (CDC), are available at: https://wwwn.cdc.gov/nchs/nhanes/Default.aspx. This survey, conducted by the CDC, aims to assess the health and nutritional status of noninstitutionalized civilians in the United States. The cohort employs a stratified, multistage probability sampling method to evaluate the health and nutritional status of

U.S. noninstitutionalized civilians. The stratified multistage probability sampling strategy assesses approximately 10,000 individuals every two years. The data collection process includes in-depth household interviews and comprehensive physical assessments. This study protocol was approved by the Institutional Review Board (IRB) of the National Center for Health Statistics, and written informed consent was obtained from all participants before completing questionnaires and collecting laboratory analysis samples, in accordance with the ethical guidelines established by the Helsinki Declaration.

2.3.2 BMI Classification

BMI is an index used to estimate body fat content in individuals of different ages and genders, calculated by dividing weight (in kilograms) by height (in meters) squared. According to the classification standards of the National Institutes of Health (NIH) and the World Health Organization (WHO), BMI is classified as follows: (1)Normal weight: BMI < 25; (2)Overweight: 25 ≤ BMI < 30; (3)Obesity Class I: 30 ≤ BMI < 35; (4)Obesity Class II: 35 ≤ BMI < 40; (5)Obesity Class III: BMI ≥ 40. These classification standards apply to White, Hispanic, and Black populations[25].

2.3.3 Epigenetic Age Acceleration

PhenoAA is measured as the residuals from a linear regression of phenotypic age on chronological age. In our previous study, PhenoAA was calculated using a composite method[14]. This algorithm integrates clinical laboratory blood chemistry parameters and various biomarkers, which were modeled through elastic net regression in NHANES III. The specific formula is as follows:

$$\mathrm{PhenoAA} = 1.435671 + \frac{\ln[-0.0059383581 \times \ln[1 - \text{mortality risk}]]}{0.08548908}$$

$$\text{mortality risk} = 1 - \mathrm{e}^{-\mathrm{e}^{\mathrm{xb}[\exp(120\mathrm{x}\gamma) - 1]/\gamma}}$$

$$\gamma = 0.007354285$$

2.3.4 Covariate Assessment

This study included multiple covariates to enable comprehensive analysis, including basic demographic information, lifestyle factors, and health status indicators.

xb = − 18.311623487 − 0.029197296 × albumin + 0.002285539 × alkaline phosphatase + 0.006379140 × creatinine + 0.177752648 × glycated hemoglobin + 0.055248172 × white blood cell count − 0.013502137 × lymphocyte percentage + 0.029331315 × mean corpuscular volume + 0.234452108 × red cell distribution width + 0.077245523 × chronological age

Demographic information includes age, gender (male or female), and race (Mexican American, non-Hispanic Black, non-Hispanic White, or other). Education level was categorized as below high school, high school, and above high school. Economic status was assessed using the poverty income ratio (PIR), with PIR < 1.3 indicating low income, 1.3-3.3 indicating middle income, 3.3-3.5 indicating upper-middle income, and > 3.5 indicating high income. Lifestyle factors include marital status (married/cohabiting, widowed/divorced/separated), smoking status (never smoked, former smoker, current smoker), alcohol consumption (never, former, light, moderate, heavy), and physical activity level was assessed using the Global Physical Activity Questionnaire (GPAQ) and categorized into quartiles (Q1, Q2, Q3, and Q4) based on weekly energy expenditure (Met-min/week). Dietary quality was assessed using the 2015 Healthy Eating Index (HEI-2015), reflecting whether an individual's diet aligns with the U.S. Department of Agriculture's Dietary Guidelines for Americans. Healthcare professionals assessed health status and included diseases such as diabetes, dyslipidemia, hypertension, and cancer.

2.4 Statistical Analysis

To determine the causal relationship between early-life BMI and accelerated epigenetic aging, we used inverse-variance weighting (IVW) as the primary method of analysis, which provides stable causal inference regardless of heterogeneity[26]. Weighted median and MR-Egger were used as secondary methods. A positive result was considered when the IVW method yielded a significant result ($p<0.05$) and when the β-values from other methods were either all positive or all negative[27]. Sensitivity analysis was performed to evaluate potential heterogeneity and horizontal pleiotropy. Cochran's Q test was used to assess heterogeneity among IVs, and MR-PRESSO was used to assess robustness of associations and to evaluate potential horizontal pleiotropy. To evaluate the sensitivity of individual SNPs, a Leave-One-Out (LOO) sensitivity test was performed. Additionally, to exclude reverse causality, bidirectional MR was used to determine the direction of the association between accelerated epigenetic aging and early-life BMI. Finally, a two-step MR method was employed to identify potential variables that accelerate or slow epigenetic aging[28].

For the NHANES dataset analysis, all statistical analyses were conducted in strict accordance with the analytical guidelines provided by the Centers for Disease Control and Prevention (CDC) to account for the complex, multistage probability sampling design of the NHANES. Appropriate multi-year Mobile Examination Center (MEC) sample weights were applied to generate nationally representative estimates. Variance and standard errors were estimated using the Taylor series linearization method, incorporating the masked variance pseudo-stratum (SDMVSTRA) and pseudo-primary sampling unit (SDMVPSU) variables. Baseline characteristics were assessed using one-way analysis of variance (ANOVA) for continuous variables and chi-square tests for categorical variables. To explore the association between BMI and PhenoAA, we used weighted generalized linear regression models, specifying a Gaussian distribution and an identity link for the continuous outcome (PhenoAA). Additionally, a subgroup analysis by gender was conducted to examine potential gender-specific effects. Both continuous BMI values and obesity categories were analyzed in the regression models to assess their impact on PhenoAA. Based on the linear regression results, we used restricted cubic splines to test for nonlinear trends among the variables. In cases of nonlinear associations, piecewise regression combined with likelihood ratio tests was used to elucidate the threshold effect of BMI on PhenoAA. Finally, to verify the impact of control variables on the relationship between BMI and PhenoAA, these variables were incorporated into an interaction effect testing model. Data processing and analysis were performed using R Studio (version 4.2.1, USA).

Multi-sample data were analyzed using GraphPad Prism 9. For normally distributed data, independent-sample t-tests or ANOVA were used; for non-normally distributed data, Mann-Whitney U tests were used for pairwise comparisons. All statistical analyses were conducted with a significance threshold of $p<0.05$. Between-group differential analyses for the proteomic and metabolomic data were conducted on $\log_2$-transformed abundances. For each prespecified contrast, multiple testing was controlled using the Benjamini-Hochberg procedure, with FDR-adjusted p-values computed across all quantified proteins and, separately, across all identified metabolites; statistical significance was defined as FDR-adjusted $p < 0.05$.

2.5 Population Study

2.5.1 Subject Recruitment

We recruited 51 overweight or obese female (age = 23.52 ± 4.88, BMI = 25.29 ± 2.97, body fat percentage = 36.76 ± 4.80) to participate in an 8-week comprehensive exercise intervention. Participants exercised three times per week for 75 minutes per session under the supervision of qualified trainers. Fasting blood samples were collected at baseline and after the completion of the 8-week exercise intervention. Concurrently, 40 male were recruited, including 21 with obesity(age = 27.57 ± 5.34, BMI = 43.29 ± 6.35) and 19 with normal weight and regular exercise habits (age = 29.42 ± 6.31, BMI = 21.96 ± 1.85). Visceral fat samples were obtained from obese metabolic weight loss surgery patients (n = 4, age = 27.5 ± 8.5, BMI = 47.13 ± 7.0), while visceral fat from men with normal weight and exercise habits was sourced from hernia surgery patients (n = 3, age = 30 ± 1.73, BMI = 24.92 ± 2.3). The research team conducted interviews with all participants to assess their exercise experience. Prior to participating in the study, all participants read and signed an informed consent form (Ethics approval number: 2021-112-K70; Clinical registration: NCT06452303）.

2.5.2 Biomarker Detection Methods

The experimental procedures involved in this study were carried out in strict accordance with the standardized protocols provided by the commercial assay kits.

Blood Sample Collection and Preparation

Before the start of the exercise intervention and after completion of the 8-week exercise program, fasting blood samples were collected from the antecubital vein. The EDTA anticoagulation tubes (Solarbio, YA1462-1pk) were stored at 4°C for at least 1 hour, followed by centrifugation (3,000 rpm for 10 minutes) to separate the plasma. Whole blood, plasma, and PBMCs were aliquoted and stored at -80°C for subsequent analyses. PBMCs were isolated from whole blood samples using density gradient centrifugation, and human peripheral blood lymphocyte separation solution (Human) was purchased from Beijing Solarbio Science & Technology Co., Ltd. (P8610-200ml).

Enzyme-Linked Immunosorbent Assay (ELISA)

Plasma levels of Glutathione and 8-OHdG were measured using commercial ELISA kits (Elabscience Biotechnology, GSH catalog number: E-EL-0026; 8-OHdG catalog number: E-EL-0028).

Real-Time qPCR

Total RNA was extracted using Trizol reagent (Invitrogen) according to the manufacturer's instructions[29]. Gene expression data were normalized to GAPDH as the housekeeping gene, and primer sequences are provided in the supplementary materials (Table 4, Supplementary File 1).

Immunohistochemistry (IHC)

IHC was performed to detect the expression of p16, p21, and p-γ-H2A.X (S139) in adipose tissue. The tissues were fixed in 4% paraformaldehyde, dehydrated, and embedded in O.C.T. Compound (Beijing Wanjing Inspirational Biotechnology, catalog number: WJ1177) for frozen sectioning. The sections were treated with 0.1% Triton X-

100 and blocked with 3% H_2O_2 to inhibit endogenous peroxidase activity, then subjected to antigen retrieval. After blocking non-specific antibody binding with BSA, the sections were incubated overnight at 4°C with primary antibodies and subsequently incubated for 1 hour at room temperature with secondary antibodies. The sections were then stained with DAB and counterstained with hematoxylin. Images were captured with an optical microscope (Olympus) and analyzed with ImageJ. The antibodies used were: p16 (ET1608-62, dilution 1:200, HUABIO), p21 (HA500156, dilution 1:200, HUABIO), p-γ-H2A.X (ET1602-2, dilution 1:100, HUABIO).

Immunofluorescence Staining（IF）

Frozen adipose tissue sections were fixed at room temperature with 4% paraformaldehyde and treated with 0.1% Triton-X 100. After blocking nonspecific antibody binding with BSA, the sections were incubated overnight at 4°C with primary antibodies, then for 1 hour at room temperature with fluorescent secondary antibodies. Images were collected using an LSM 780 confocal microscope (Zeiss) and analyzed in ImageJ. The antibodies used were: p16 (ET1608-62, dilution 1:200, HUABIO), p21 (HA500156, dilution 1:200, HUABIO), p-γ-H2A.X (R010101, dilution 1:100, Epizyme).

Metabolomics

A 100 μL plasma sample was added to a 1.5 mL centrifuge tube, followed by the addition of 400 μL extraction solution (acetonitrile: methanol = 1:1) containing 0.02 mg/mL internal standard (L-2-chlorophenylalanine). The mixture was vortexed for 30 seconds and subjected to low-temperature ultrasound extraction for 30 minutes (5°C, 40 KHz). The sample was then left at -20°C for 30 minutes, followed by centrifugation at 13,000 g for 15 minutes at 4°C. The supernatant was collected, dried under nitrogen, and reconstituted in 100 μL of solution (acetonitrile: water = 1:1). The reconstituted sample was sonicated for 5 minutes (5°C, 40 KHz), and centrifuged again at 13,000 g for 10 minutes at 4°C. The supernatant was transferred to an injection vial with an insert for analysis. Detailed methods are described in previous studies[30]. Metabolomics was performed by Shanghai Majorbio Bio-Pharm Technology Co., Ltd.

Proteomics

A 100 μL plasma sample was added to an appropriate amount of pre-chilled protein lysis buffer (8 M urea + 1% SDS, containing protease inhibitors). The sample was subjected to non-contact low-temperature sonication for 30 minutes. After centrifugation at 14,000 g for 15 minutes at 8°C, the supernatant was collected. Protein concentration was measured using the BCA method, following the instructions provided with the BCA reagent. After protein quantification, SDS-PAGE electrophoresis was performed. Detailed methods can be found in previous studies[31]. Proteomics was performed by Shanghai Majorbio Bio-Pharm Technology Co., Ltd.

3 Results

3.1 MR Estimates

Genetically predicted early-life BMI shows a causal association with increased epigenetic age acceleration across multiple clocks (Figure 2). Specifically, early-life BMI was positively associated with IEAA greater acceleration (β =0.231, 95%Cl 0.001 to 0.461; p=4.92×10^{-2}); GrimAA (β =0.369, 95%Cl 0.157 to 0.581; p=6.52×10^{-4}); HannumAA (β =0.267, 95%Cl 0.045 to 0.49; p=1.85×10^{-2}); and PhenoAA(β =0.560, 95%Cl 0.237 to 0.775; p=2.31×10^{-4}). Childhood obesity was likewise positively associated with IEAA greater acceleration (β =0.162, 95%Cl 0.033 to 0.291; p=1.36×10^{-3}); GrimAA(β =0.164, 95%Cl 0.043 to 0.285; p=7.71×10^{-3}); HannumAA (β =0.139, 95%Cl 0.035 to 0.244; p=8.90×10^{-3}); and PhenoAA(β =0.231, 95%Cl 0.097 to 0.365; p=7.20×10^{-4}). Subcutaneous adiposity was associated with GrimAA (β =0.471, 95% CI 0.169 to 0.773; p=2.26×10^{-3}). Together, these findings indicate that higher early-life adiposity is associated with accelerated epigenetic ageing.

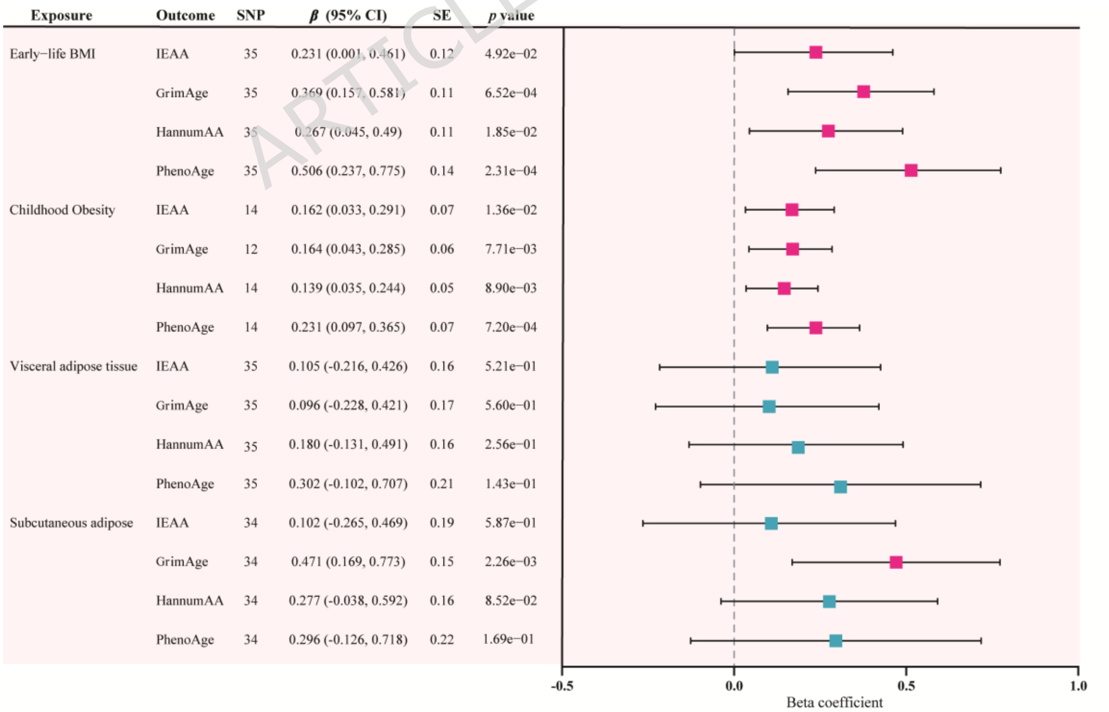

Figure 2. Causal associations between early-life weight-related exposures and epigenetic age acceleration. Forest plots present inverse-variance weighted Mendelian

randomization estimates for the associations of early-life BMI, childhood obesity, visceral adipose tissue, and subcutaneous adipose tissue with IEAA, GrimAge acceleration, Hannum age acceleration, and PhenoAge acceleration. Squares indicate β estimates and horizontal lines indicate 95% confidence intervals (CIs). The dashed vertical line represents the null effect (β = 0). The left panel shows the number of SNPs, β (95% CI), SE, and p value.

3.2 MR Mediation Analysis

Using a two-step MR framework, we assessed PA as a mediator between early-life BMI and epigenetic age acceleration (Supplementary Table 1). PA showed partial mediation. Genetically predicted PA was associated with slower ageing across four clocks, with direct effects of IEAA (β = −0.06, 95% CI −0.109 to −0.001), GrimAA (β = −0.09, 95% CI −0.138 to 0.039), HannumAA (β = −0.10, 95% CI −0.151 to −0.041), and PhenoAA (β = −0.12, 95% CI −0.183 to −0.058). The proportions of the early-life BMI-epigenetic ageing association mediated by PA were 10.74%, 10.88%, 16.22%, and 10.70%, respectively. Supplementary MVMR analyses further suggested a potential causal relationship between early-life BMI and PA (Supplementary Figure 1), indicating potential interrelatedness between these exposures. This finding is consistent with the mediation analysis and supports consideration of PA as a relevant factor in the association between early-life BMI and epigenetic ageing.

3.3 Horizontal Pleiotropy and Sensitivity Analysis

To assess the robustness of the statistical results, sensitivity analysis was conducted for the four exposures and accelerated epigenetic aging (Table 5, Supplementary File 1). The MR-Egger intercept test indicated no horizontal pleiotropy (p > 0.05), and no outliers or heterogeneity were observed (p > 0.05). Furthermore, the Leave-One-Out analysis and funnel plot indicated no violations in the estimates (Supplementary File 1), confirming the robustness and reliability of the statistical results.

3.4 Demographic Characteristics of NHANES

This study included baseline data from 20,337 adults, classified by BMI categories. As BMI increased, the distribution of accelerated epigenetic age shifted towards higher values. Significant differences were observed across demographic indicators (age, gender, race), lifestyle factors (marital status, education, PIR, smoking status, alcohol consumption, HEI score), and health status indicators (diabetes, hypertension, hyperlipidemia). Among participants with accelerated epigenetic aging, a higher proportion were male. Non-Hispanic Whites made up the largest proportion of the Obesity group, which also had poorer economic status and a higher prevalence of cardiovascular and metabolic diseases (Table 3, Supplementary File 1).

3.5 Association analysis between BMI and PhenoAA in American adults

A significant positive association was observed between BMI and PhenoAA. In the fully adjusted model, compared to the group with BMI < 25, overweight individuals exhibited accelerated PhenoAA [Overweight: $\beta = 0.83$, 95% CI (0.66, 1.00), $p < 0.0001$]. When BMI ≥ 30, the acceleration of PhenoAA increased significantly, and aging acceleration increased progressively with higher BMI [Obesity I: $\beta = 1.73$, 95% CI (1.54, 1.93), $p < 0.0001$; Obesity II: $\beta = 2.93$, 95% CI (2.67, 3.20), $p < 0.0001$; Obesity III: $\beta = 4.64$, 95% CI (4.30, 4.99), $p < 0.0001$]. The restricted cubic spline (RCS) analysis revealed a nonlinear threshold for BMI at 27.37 (p for Nonlinear = 0.005) (Figure 3a-b).

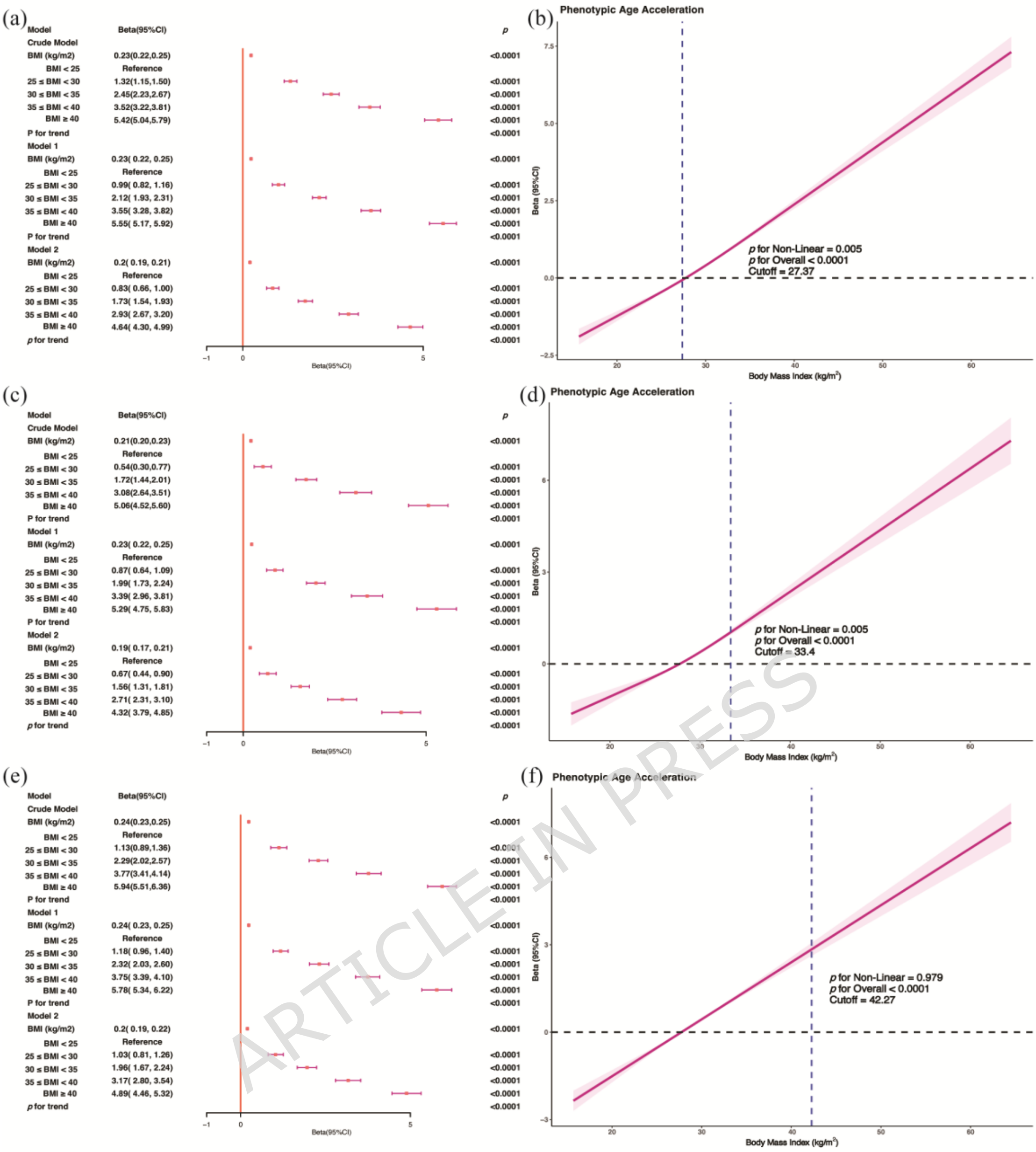

Figure 3.（a-b）Male and Female, (c-d)Male; (e-f)Female. Association analysis between BMI and PhenoAA in American adults Legend: Crude Model is the unadjusted model. Model 1 adjusted for age, sex, race,PIR, education, BMI, marital status, smoke, alcohol, HEI. Model 2 adjusted for age, sex, race, PIR, education, BMI, marital status, smoke, alcohol, HEI, diabetes, hypertension, hyperlipidemia and cancer.This study used a multivariate linear regression model to analyze the relationship between BMI and PhenoAA in American adults. The results are shown in Figure 3 (a-f). Figure 3a,3c,3e the Y-axis is labeled "Beta (95% CI)"; Figure 3b, 3d, and 3f, the X-axis is labeled "Body Mass Index (kg/m²)", and the Y-axis is labeled "Beta (95% CI)".

Subgroup analysis by gender yielded consistent results. In male, after becoming overweight, individuals also showed accelerated PhenoAA, and this acceleration increased with higher BMI [Overweight: β = 0.67, 95% CI (0.44, 0.90), $p < 0.0001$; Obesity I: β = 1.56, 95% CI (1.31, 1.81), $p < 0.0001$; Obesity II: β = 2.71, 95% CI (2.31, 3.10), $p < 0.0001$; Obesity III: β = 4.32, 95% CI (3.79, 4.85), $p < 0.0001$]. RCS analysis indicated a nonlinear threshold for BMI at 33.4 (p for Nonlinear = 0.005) (Figure 3c-d).

In the female population, the degree of PhenoAA acceleration was more pronounced than in male. Compared to female with a BMI < 25, both overweight and obese female exhibited significantly accelerated PhenoAA aging. Specifically, overweight female had a PhenoAA acceleration of β = 1.03, 95% CI (0.81, 1.26), $p < 0.0001$, while obese female showed progressively greater acceleration of PhenoAA [Obesity I: β = 1.96, 95% CI (1.67, 2.24), $p < 0.0001$; Obesity II: β = 3.17, 95% CI (2.80, 3.54), $p < 0.0001$; Obesity III: β = 4.89, 95% CI (4.46, 5.32), $p < 0.0001$]. RCS analysis revealed a linear threshold for BMI at 42.27 (p for Nonlinear = 0.979) (Figure 3e-f).

3.6 Threshold Effect Analysis of the Relationship Between BMI and PhenoAA

This study demonstrated a nonlinear relationship between BMI levels and PhenoAA. Specifically, when BMI < 27.37, the relationship between BMI and PhenoAA was significant [β = 0.149, 95% CI (0.125, 0.173), $p < 0.001$]. However, when BMI ≥ 27.37, the relationship between the two changed [β = 0.198, 95% CI (0.186, 0.210), $p < 0.001$]. The likelihood ratio test showed statistical significance ($p < 0.01$), providing strong evidence for the Two-piecewise linear regression model (Table 1).

In the male subgroup, when BMI < 33.4, the relationship between BMI and PhenoAA was significant [β = 0.154, 95% CI (0.134, 0.173), $p < 0.001$]. However, when BMI ≥ 33.4, the relationship changed [β = 0.218, 95% CI (0.188, 0.248), $p < 0.001$]. The likelihood ratio test ($p < 0.01$) indicated that the two-piecewise linear model provided a significantly better fit than the single linear model (Table 1).

For females, when BMI < 42.27, the relationship between BMI and PhenoAA was significant [β = 0.207, 95% CI (0.194, 0.221), $p < 0.001$]. However, when BMI ≥ 42.27, the relationship changed [β = 0.121, 95% CI (0.073, 0.169), $p < 0.001$]. The likelihood ratio test ($p < 0.01$) indicated that the two-piecewise linear model provided a significantly better fit than the single linear model (Table 1).

These results highlight the importance of considering the nonlinear threshold effect of BMI levels in predicting PhenoAA, indicating that the acceleration of aging in both male and female is influenced by specific BMI thresholds.

Table 1.Threshold effect analysis of the relationship between BMI and PhenoAA

Outcome	Beta	95%CI	*p*-value
BMI(All)			
Two-piecewise linear regression model			
Inflection point	27.37		
BMI<27.37	0.149	0.125，0.173	<0.001
BMI≥27.37	0.198	0.186,0.210	<0.001
p for Log-likelihood ratio test			0.002
BMI(Male)			
Two-piecewise linear regression model			
Inflection point	33.4		
BMI<33.4	0.154	0.134,0.173	<0.001
BMI≥33.4	0.218	0.188,0.248	<0.001
p for Log-likelihood ratio test			0.003
BMI(Female)			
Two-piecewise linear regression model			
Inflection point	42.27		
BMI<42.27	0.207	0.194,0.221	<0.001
BMI≥42.27	0.121	0.073,0.169	<0.001
p for Log-likelihood ratio test			0.002

Notes: A two-piecewise linear regression model was used to assess the threshold effect of BMI on PhenoAA. The inflection point denotes the estimated BMI threshold. Regression coefficients (Beta), 95% confidence intervals (95% CI) and p-values are reported separately for BMI below and above the inflection point. The log-likelihood ratio test compares the two-piecewise model with a single linear model; $p < 0.05$ indicates a significantly better fit of the two-piecewise model, supporting the presence of an inflection threshold.

3.7 Interaction Effect Test Between BMI and PhenoAA

This study analyzed the relationship between BMI and PhenoAA and assessed the interactions among gender, age, race, marital status, education level, PIR, smoking status, alcohol consumption, HEI, diabetes, hyperlipidemia, hypertension, and cancer

(Supplementary Materials File 2). The results indicated that age, race, hypertension, and cancer had statistically significant interaction effects with BMI and PhenoAA ($p < 0.05$). This suggests that the relationship between BMI and PhenoAA may be influenced by other factors across subgroups, including age, race, and health status.

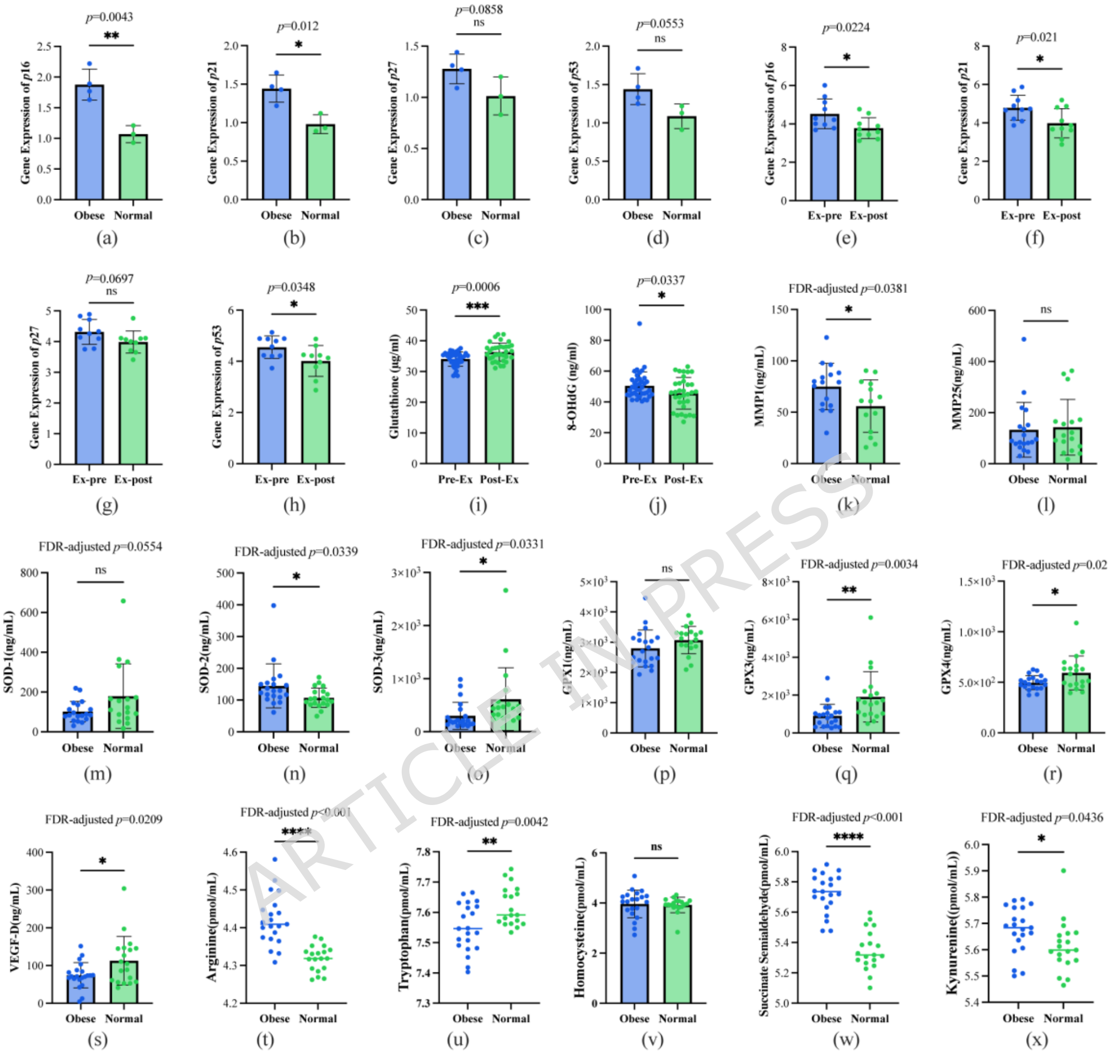

Figure 4. Multi-sample comparison of metabolic, inflammatory, and oxidative stress biomarkers. a-d: Detection of cell cycle regulatory gene expression in adipose tissue of the third group of subjects; e-j: Detection results of aging-related markers in plasma and PBMCs of the first group of subjects; k-x: Detection of oxidative stress markers in plasma metabolites and proteomics of the second group of subject. *p<0.05,**p<0.01,***p<0.0001.

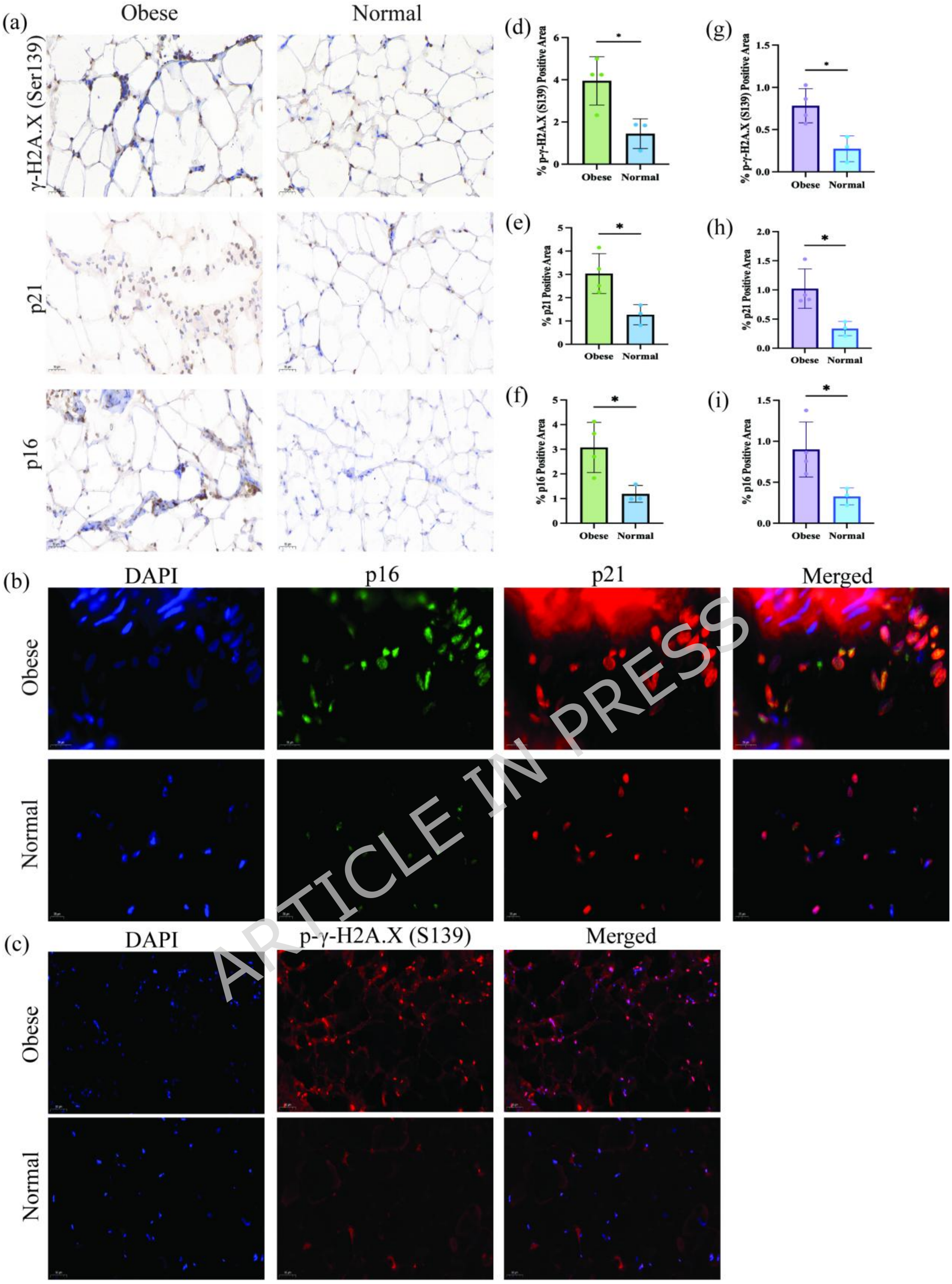

Figure 5. (a) Immunohistochemical staining of senescence markers in visceral adipose tissue; (b-c) Immunofluorescence staining of visceral adipose tissue; (d-f) Statistical analysis of positive cell area from immunohistochemical staining; (g-i) Statistical analysis of fluorescence intensity from immunofluorescence staining. *$p<0.05$.

3.8 Identification of Aging Biomarker Associated with Physical Activities

Obesity Accelerates Aging and DNA Damage

Compared to individuals with normal weight and regular exercise habits, obese individuals showed a significant upregulation in the expression of cell cycle regulation-related genes (p16 and p21) in their omental fat tissue (Figure 4. a-d). Additionally, IHC further supported these findings. Compared with the normal-weight group, adipose tissue from obese individuals showed a markedly higher abundance of p16, p21, and phospho-γ-H2AX (Ser139)-positive cells, indicating increased cellular senescence and DNA damage in obesity (Figure 5a, d-f). Consistent with these results, IF staining further verified these changes and revealed that the signals were predominantly localized to the nuclei (Figure 5b-c, g-i). The obese group also exhibited stronger nuclear fluorescence intensity, providing additional histological evidence that obesity is associated with accelerated senescence-related alterations and increased DNA damage in adipose tissue.

Physical Activity Reduces DNA Damage and Oxidative Stress

In the first group of overweight or obese participants who underwent an 8-week comprehensive exercise intervention, we observed a significant reduction in the expression levels of p16, p21, and p53 genes in peripheral blood mononuclear cells, indicating that exercise effectively mitigated the upregulation of aging-related genes (Figure 4.e-j). Additionally, plasma glutathione levels increased, and 8-OHdG levels decreased, further supporting the potential role of exercise in reducing oxidative stress and DNA damage (Figure 4.f). In the second group of participants, we collected their fasting plasma and conducted metabolomics and proteomics analyses (Figure 4.k-x). The results showed that the plasma levels of MMP1, SOD-2, Arginine, Succinate Semialdehyde, and Kynurenine were higher in the obese group, suggesting significant oxidative stress and metabolic disturbances in this population. In contrast, in the healthy group with normal weight and regular exercise habits, the plasma levels of SOD-3, GPX3, GPX4, VEGF-D, and Tryptophan were higher, indicating that maintaining normal weight and good exercise habits helps regulate antioxidant capacity and promote healthy metabolism.

4. Discussion

Prior research has been largely associative and focused mainly on adult populations. By integrating Mendelian randomization, population-based analyses spanning childhood and adulthood, and molecular profiling of biospecimens collected in the context of exercise intervention, the present study provides multi-level evidence on the relationships among early-life BMI, physical activity, and biological aging. Our findings suggest that higher early-life BMI is associated with accelerated epigenetic aging across multiple clocks. Physical activity may partially attenuate this adverse association. However, the potentially mediating role of physical activity should be interpreted cautiously, as MVMR is designed to estimate the independent effects of correlated exposures rather than to directly establish mediation. In the NHANES analysis, higher adult BMI was likewise positively associated with PhenoAge acceleration, with progressively greater advancement observed across BMI categories. Compared with normal-weight individuals (BMI < 25 kg/m^2), PhenoAA was 0.83 years higher in the overweight group, 1.73 years higher in class I obesity, 2.93 years higher in class II obesity, and 4.64 years higher in class III obesity, indicating a clear graded relationship across BMI-defined adiposity categories. In addition, analyses of blood and adipose biospecimens suggested that obesity-related metabolic burden was accompanied by greater DNA damage, oxidative stress, and aging-related molecular alterations. In contrast, regular physical activity and maintaining a healthier weight were associated with more favorable aging-related profiles. Taken together, these results support a link between excess adiposity and accelerated biological aging across the life course and further suggest that physical activity may help mitigate this process. These findings provide supportive evidence for lifestyle-based strategies to promote healthier aging.

Biological aging is not merely a reflection of chronological time; rather, its molecular hallmarks often emerge subtly during early life, long before they manifest clinically. A study on preschool children (ages 3-4) found that, compared to their normal-weight peers, obese children exhibited significantly shorter telomere lengths in peripheral blood leukocytes, suggesting that changes in biomarkers associated with aging may appear early in life[32]. Furthermore, a prospective cohort study involving children aged 8-14 confirmed that higher BMI during childhood was significantly associated with accelerated epigenetic aging[33]. Together, these findings indicate that the onset of aging in obese individuals begins much earlier than the established physiological age. Without intervention, the acceleration of aging in early life may persist into adulthood. We found that PhenoAA was particularly sensitive to early-life BMI. As an “extrinsic” clock, PhenoAA reflects not only cell-intrinsic methylation ageing but also shifts in blood-cell composition and the broader metabolic milieu[34]. It was explicitly developed to approximate phenotypic age, trained on mortality and clinical risk markers, with CpGs selected accordingly, so obesity-related chronic inflammation and metabolic dysregulation are more readily captured as PhenoAA acceleration[35]. We also observed that, among individuals with obesity, female exhibited higher epigenetic age than male. This pattern likely relates to alterations in the sex-steroid milieu; oestrogen has antioxidant, anti-inflammatory, and pro-repair effects[36], whereas obesity-associated hyperandrogenism in female can suppress oestrogen biosynthesis and potentially hasten biological ageing[37]. Consistent with this, large studies report that earlier menopause is linked to faster epigenetic ageing, with each year of earlier menopause associated with a further advance in DNA methylation age[38]. Notably, we did not observe a clear causal association between visceral adipose tissue (VAT) or abdominal subcutaneous adipose tissue (ASAT) volume and epigenetic aging measures. This null finding may partly reflect the characteristics of the imaging cohort. Although the mean BMI ranged from 26.5 to 27.4 kg/m^2, indicating an overall overweight population, VAT and ASAT are imaging-derived quantitative traits for which the available genetic instruments typically explain only a limited proportion of variance[22]. In addition, the relatively small sample size further reduced statistical power to detect potential effects of fat distribution on accelerated aging. These findings also suggest that the aging-related risk associated with higher BMI in early life may not be entirely irreversible. Sustained improvement in body weight and metabolic health during adulthood may help attenuate epigenetic age acceleration; however, this life-course effect of weight trajectory still requires confirmation in longitudinal studies with follow-up data.

Obesity-associated accelerated aging is not confined to a single organ or tissue; rather, it can involve multiple systems, including the liver, omental adipose tissue, and the brain[8][9][39]. Experimental and mechanistic studies in cellular and animal models suggest that excessive lipid accumulation may trigger chronic low-grade inflammation, oxidative stress, and metabolic dysfunction, thereby promoting the earlier onset of molecular damage and functional decline[40]. Emerging evidence further indicates that obesity-related aging is not confined to a single tissue but rather involves coordinated alterations across multiple organs and biological systems, including adipose tissue dysfunction, stress-response pathways, and systemic inflammatory remodeling[41][42]. In contrast, physical activity is widely recognized as a scalable intervention with systemic benefits. It may help attenuate obesity- or metabolic stress-related acceleration of biological aging, which is consistent with our findings[43][44]. Mechanistic studies further suggest that exercise exerts anti-aging effects through multi-organ processes, including the reduction of chronic inflammation, improvement of metabolic homeostasis, enhancement of mitochondrial function, and modulation of inter-organ communication. These beneficial adaptations have been observed across several tissues, including adipose tissue, the liver, skeletal muscle, and the central nervous system, underscoring the broad biological relevance of physical activity in regulating aging[45][46][47]. Taken together, the protective effects of physical activity on biological aging are more likely to reflect integrated whole-body adaptations than a single organ-specific mechanism.

We employed an integrated design across diverse populations, improving representativeness and generalizability. Collectively, our findings showed that obesity accelerates aging, whereas physical activity consistently mitigates it, providing a scientific basis for exercise-based strategies to promote healthy aging. Because aging is a multisystem process, drugs that act on single targets often yield limited system-wide benefits. By contrast, exercise elicits coordinated, cross-organ adaptations with robust anti-aging effects. Elevated BMI in early life may signal a trajectory toward accelerated biological aging, underscoring childhood and adolescence as a critical window for preventive intervention. Health education and early school- and community-based programs that promote a healthy diet and regular physical activity may therefore help reduce obesity, prevent adult chronic disease, and support healthy aging across the life course. To translate these life-course strategies into actionable prevention and clinical management, it is essential to have a scalable biomarker that quantifies biological aging and tracks intervention response. Epigenetic clocks provide such a quantitative readout by capturing cumulative biological burden beyond chronological age and conventional adiposity indices, enabling risk stratification and identifying individuals with disproportionate aging who may benefit from earlier or more intensive prevention and clinical management. Importantly, the potential reversibility of epigenetic modifications does not diminish clinical utility; rather, it makes epigenetic age an actionable marker. Epigenetic clocks can serve as intermediate phenotypes or endpoints to evaluate whether interventions (exercise training, dietary programs, or metabolic surgery) achieve system-wide aging benefits and to support longitudinal monitoring and tailoring of intervention intensity over time.

Limitations and Future Prospects

Several limitations merit consideration. First, obesity in the public datasets was primarily defined by BMI, which is not a gold-standard measure of adiposity and cannot distinguish fat and lean mass. This may have introduced misclassification and reduced estimate precision. In addition, BMI reflects overall adiposity but does not fully capture the biological heterogeneity of pathological obesity, limiting the specificity of our causal interpretation. Future studies should incorporate more refined adiposity-related phenotypes and methods such as multivariable or pathway-specific MR. Second, caution is needed when comparing the MR and NHANES findings. The MR analysis evaluated the effect of early-life adiposity on epigenetic aging. In contrast, the NHANES analysis was limited to adults because data required to calculate epigenetic aging measures in minors were unavailable. As such, the NHANES analysis does not constitute a strictly matched validation, but rather provides complementary observational evidence for the association between obesity and accelerated aging in adulthood. Third, although adipose tissue and blood sample findings supported the causal and cohort-based results, the sample sizes were modest and require confirmation in larger studies. Finally, differences in age structure between the GWAS population used for the MR analysis and the NHANES sample may have contributed to differences in association magnitude, limiting direct comparison of effect estimates across datasets. Therefore, consistency in the direction of findings may be more informative than exact similarity in effect sizes.

Abbreviations

IEAA Intrinsic Epigenetic Age Acceleration

GrimAA GrimAge Acceleration

HannumAA Hannum Age Acceleration

PhenoAA PhenoAge Acceleration

BMI Body mass index

MR Mendelian randomization

SNPs single-nucleotide polymorphisms

IVs instrumental variables

NHANES National Health and Nutrition Examination Survey

CDC Centers for Disease Control and Prevention

IRB Institutional Review Board

NIH National Institutes of Health

WHO World Health Organization

GPAQ Global Physical Activity Questionnaire

HEI-2015 2015 Healthy Eating Index

IVW Inverse-variance weighting

LOO Leave-One-Out

ANOVA One-way analysis of variance

ELISA Enzyme-Linked Immunosorbent Assay

IHC Immunohistochemistry

IF Immunofluorescence

VAT Visceral adipose tissue

ASAT Abdominal subcutaneous adipose tissue

GR Glucocorticoid-glucocorticoid receptor

GREs Glucocorticoid response elements

BBB Blood-brain barrier

Acknowledgements

We would like to express our sincere gratitude to the researchers and teams who have made their valuable research data publicly available.

Ethics approval and consent to participate

All the cohorts and samples involved in the study adhered to national and international guidelines, including the Deontological Code and the Declaration of Helsinki, and complied with current personal data protection regulations. The research using publicly available data was approved by the relevant authorities, with all participants having provided informed consent. For the multi-biomarker studies, ethical approval was obtained from China-Japan Friendship Hospital and Beijing Sport University, and all participants signed informed consent forms (Ethics approval number: 2021-112-K70).

Consent

Written informed consent for publication was obtained from every patient; for minors, consent was secured from their parents or legal guardians.

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Author Contributions: Dr. Zhou was responsible for the Mendelian randomization data analysis and adipose tissue sample collection, Dr. Wu was responsible for the Nhanes cohort data acquisition and analysis, and Dr. Zhou and Master Chen were responsible for laboratory testing. The three are jointly responsible for the authenticity of the data and the accuracy of the data analysis; Concept design: Bojun, Dongzhe, Biao, Lei, Hua; Acquisition, analysis or interpretation of data: Bojun, Dongzhe; Drafting the manuscript: Bojun, Dongzhe, Biao, Lei, Hua; Laboratory testing: Bojun, Shuting, Jiaming, Shijie, Ninghao; Subject recruitment and adipose sample acquisition: Xiaochuan, Shiyan, Bojun, Biao; Statistical analysis: Bojun, Dongzhe, Lianghao, Lei

Conflicts of interest disclosure

All authors have no conflicts of interest or financial ties to disclose.

Research registration unique identifying number(UIN)

Name of the registry: China-Japan Friendship Hospital National Integrated Chinese and Western Medicine Medical Center

Unique identifying number or registration ID: NCT06452303.

Hyperlink to your specific registration (must be publicly accessible and will be checked):https://www.centerwatch.com/clinical-trials/listings/NCT06452303/the-patient-cohort-for-bariatric-surgery

Data availability statement

For access to multi-sample data or analysis code, please contact Bojun Zhou (phdzhou@qq.com). The Nhanes data and Mendelian randomization data can be accessed directly at https://wwwn.cdc.gov/nchs/nhanes/Default.aspx and https://gwas.mrcieu.ac.uk/ respectively.

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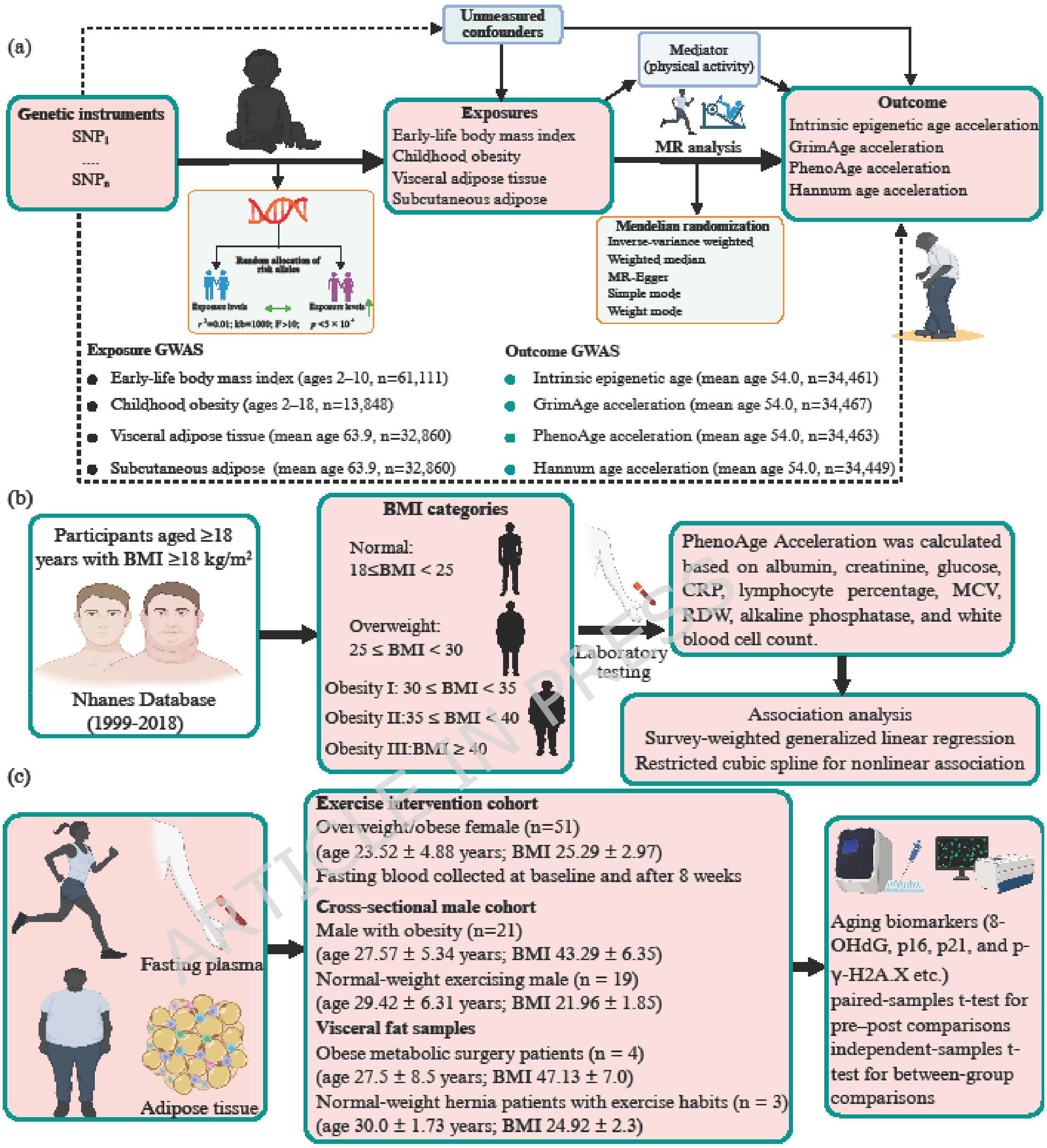
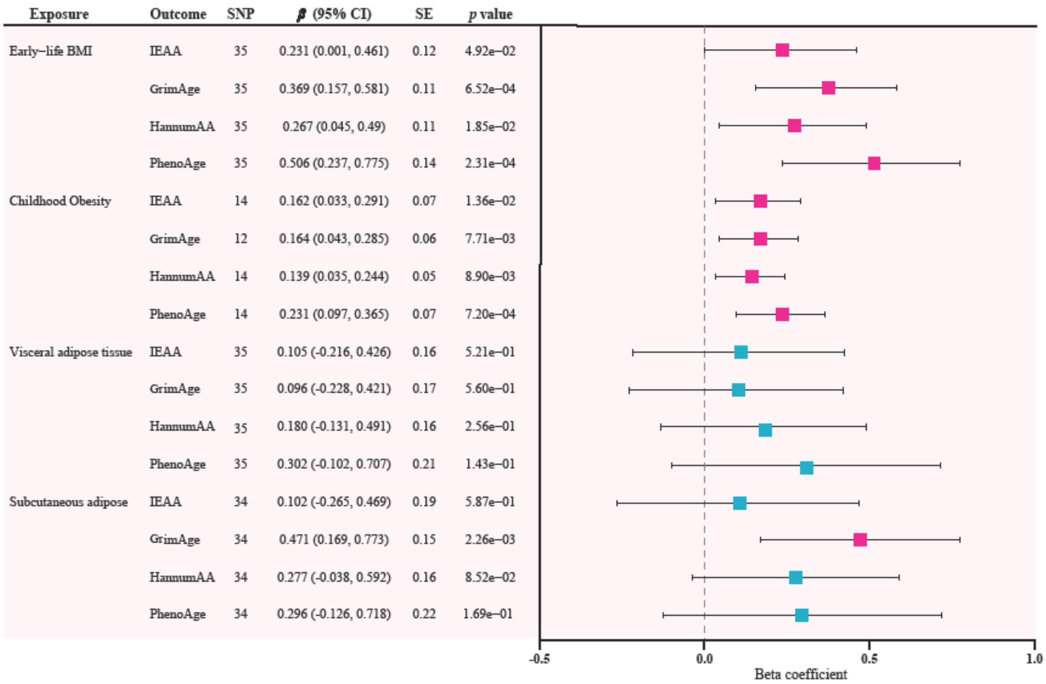
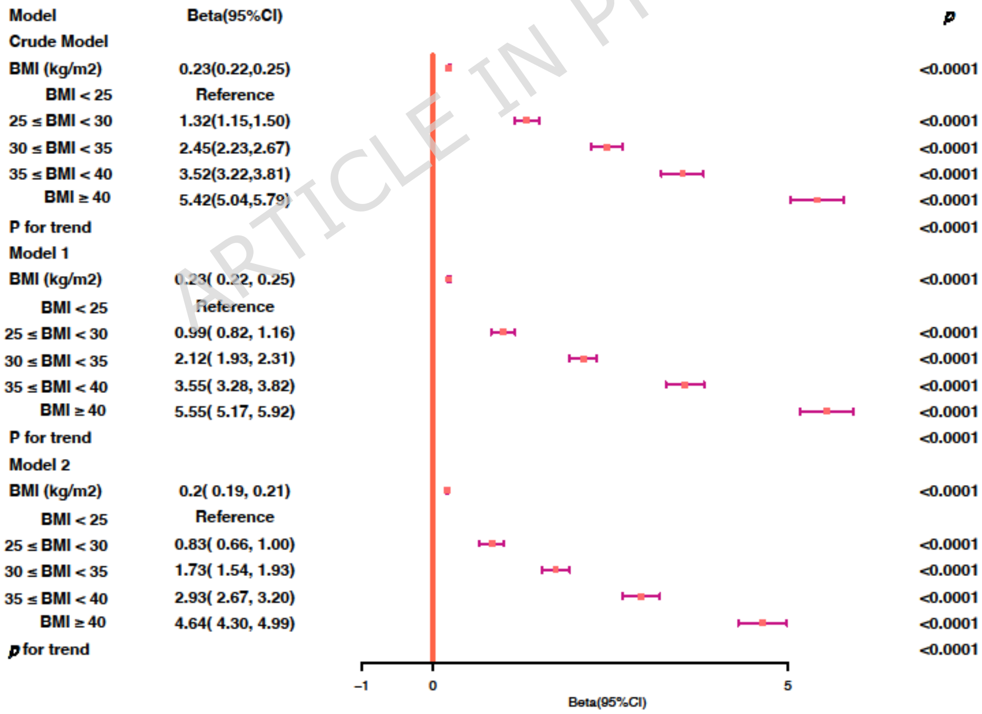
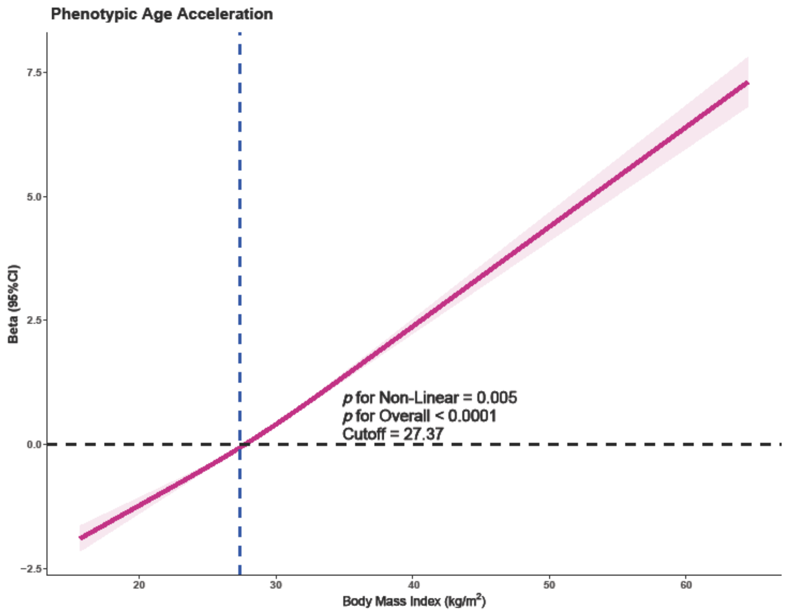

(a)
Unmeasured confounders
Mediator (physical activity)
Genetic instruments
SNP$_1$
....
SNP$_n$
Exposures
Early-life body mass index
Childhood obesity
Visceral adipose tissue
Subcutaneous adipose
MR analysis
Outcome
Intrinsic epigenetic age acceleration
GrimAge acceleration
PhenoAge acceleration
Hannum age acceleration
Random allocation of risk alleles
Exposure levels
Exposure levels
Mendelian randomization
Inverse-variance weighted
Weighted median
MR-Egger
Simple mode
Weight mode
Exposure GWAS
Early-life body mass index (ages 2–10, n=61,111)
Childhood obesity (ages 2–18, n=13,848)
Visceral adipose tissue (mean age 63.9, n=32,860)
Subcutaneous adipose (mean age 63.9, n=32,860)
Outcome GWAS
Intrinsic epigenetic age (mean age 54.0, n=34,461)
GrimAge acceleration (mean age 54.0, n=34,467)
PhenoAge acceleration (mean age 54.0, n=34,463)
Hannum age acceleration (mean age 54.0, n=34,449)
(b)
Participants aged ≥18 years with BMI ≥18 kg/m^2
Nhanes Database (1999-2018)
BMI categories
Normal: 18≤BMI < 25
Overweight: 25 ≤ BMI < 30
Obesity I: 30 ≤ BMI < 35
Obesity II:35 ≤ BMI < 40
Obesity III:BMI ≥ 40
Laboratory testing
PhenoAge Acceleration was calculated based on albumin, creatinine, glucose, CRP, lymphocyte percentage, MCV, RDW, alkaline phosphatase, and white blood cell count.
Association analysis
Survey-weighted generalized linear regression
Restricted cubic spline for nonlinear association
(c)
Fasting plasma
Adipose tissue
Exercise intervention cohort
Overweight/obese female (n=51)
(age 23.52 ± 4.88 years; BMI 25.29 ± 2.97)
Fasting blood collected at baseline and after 8 weeks
Cross-sectional male cohort
Male with obesity (n=21)
(age 27.57 ± 5.34 years; BMI 43.29 ± 6.35)
Normal-weight exercising male (n = 19)
(age 29.42 ± 6.31 years; BMI 21.96 ± 1.85)
Visceral fat samples
Obese metabolic surgery patients (n = 4)
(age 27.5 ± 8.5 years; BMI 47.13 ± 7.0)
Normal-weight hernia patients with exercise habits (n = 3)
(age 30.0 ± 1.73 years; BMI 24.92 ± 2.3)
Aging biomarkers (8-OHdG, p16, p21, and p-γ-H2A.X etc.)
paired-samples t-test for pre–post comparisons
independent-samples t-test for between-group comparisons



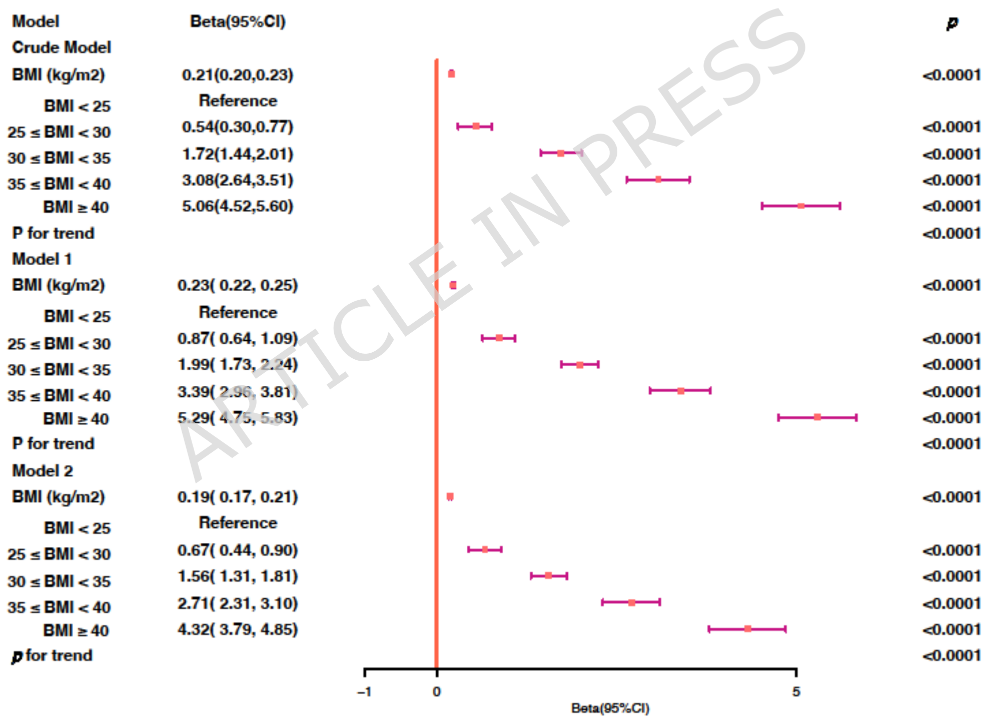
Exposure Outcome SNP β (95% CI) SE p value
Early−life BMI IEAA 35 0.231 (0.001, 0.461) 0.12 4.92e−02
GrimAge 35 0.369 (0.157, 0.581) 0.11 6.52e−04
HannumAA 35 0.267 (0.045, 0.49) 0.11 1.85e−02
PhenoAge 35 0.506 (0.237, 0.775) 0.14 2.31e−04
Childhood Obesity IEAA 14 0.162 (0.033, 0.291) 0.07 1.36e−02
GrimAge 12 0.164 (0.043, 0.285) 0.06 7.71e−03
HannumAA 14 0.139 (0.035, 0.244) 0.05 8.90e−03
PhenoAge 14 0.231 (0.097, 0.365) 0.07 7.20e−04
Visceral adipose tissue IEAA 35 0.105 (-0.216, 0.426) 0.16 5.21e−01
GrimAge 35 0.096 (-0.228, 0.421) 0.17 5.60e−01
HannumAA 35 0.180 (-0.131, 0.491) 0.16 2.56e−01
PhenoAge 35 0.302 (-0.102, 0.707) 0.21 1.43e−01
Subcutaneous adipose IEAA 34 0.102 (-0.265, 0.469) 0.19 5.87e−01
GrimAge 34 0.471 (0.169, 0.773) 0.15 2.26e−03
HannumAA 34 0.277 (-0.038, 0.592) 0.16 8.52e−02
PhenoAge 34 0.296 (-0.126, 0.718) 0.22 1.69e−01
-0.5
0.0
0.5
1.0
Beta coefficient



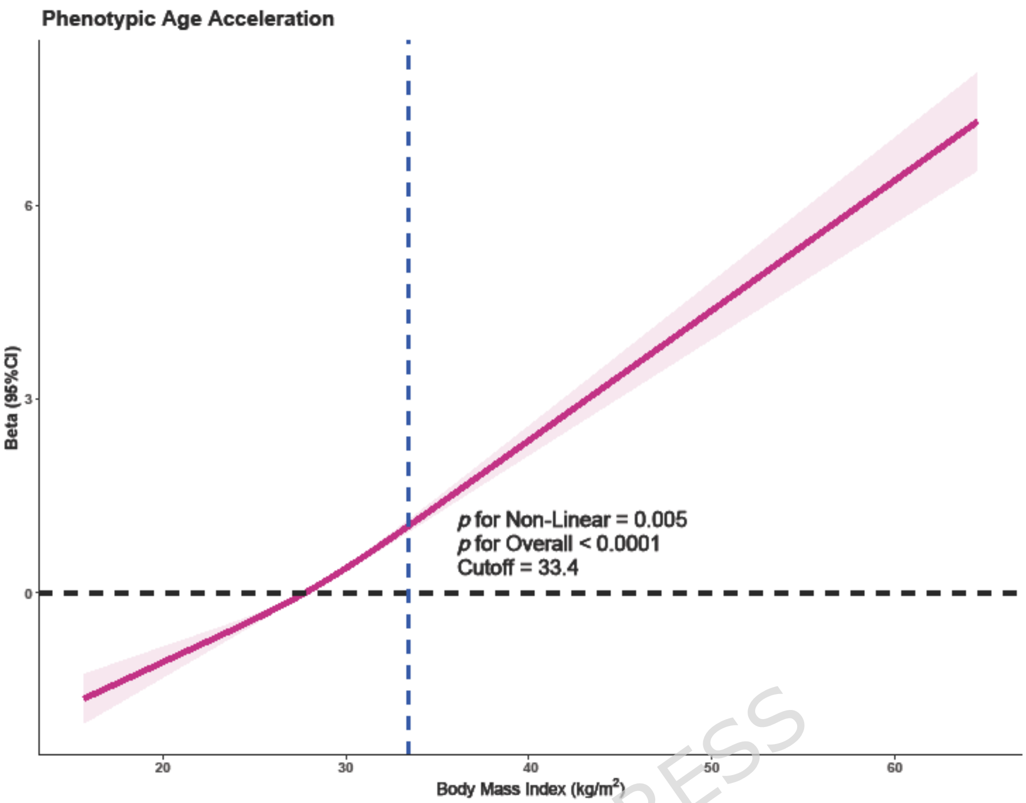
Model Beta(95%CI) p
Crude Model
BMI (kg/m2) 0.23(0.22,0.25) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 1.32(1.15,1.50) <0.0001
30 ≤ BMI < 35 2.45(2.23,2.67) <0.0001
35 ≤ BMI < 40 3.52(3.22,3.81) <0.0001
BMI ≥ 40 5.42(5.04,5.79) <0.0001
P for trend <0.0001
Model 1
BMI (kg/m2) 0.23(0.22, 0.25) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 0.99(0.82, 1.16) <0.0001
30 ≤ BMI < 35 2.12(1.93, 2.31) <0.0001
35 ≤ BMI < 40 3.55(3.28, 3.82) <0.0001
BMI ≥ 40 5.55(5.17, 5.92) <0.0001
P for trend <0.0001
Model 2
BMI (kg/m2) 0.2(0.19, 0.21) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 0.83(0.66, 1.00) <0.0001
30 ≤ BMI < 35 1.73(1.54, 1.93) <0.0001
35 ≤ BMI < 40 2.93(2.67, 3.20) <0.0001
BMI ≥ 40 4.64(4.30, 4.99) <0.0001
p for trend <0.0001
−1
0
5
Beta(95%CI)

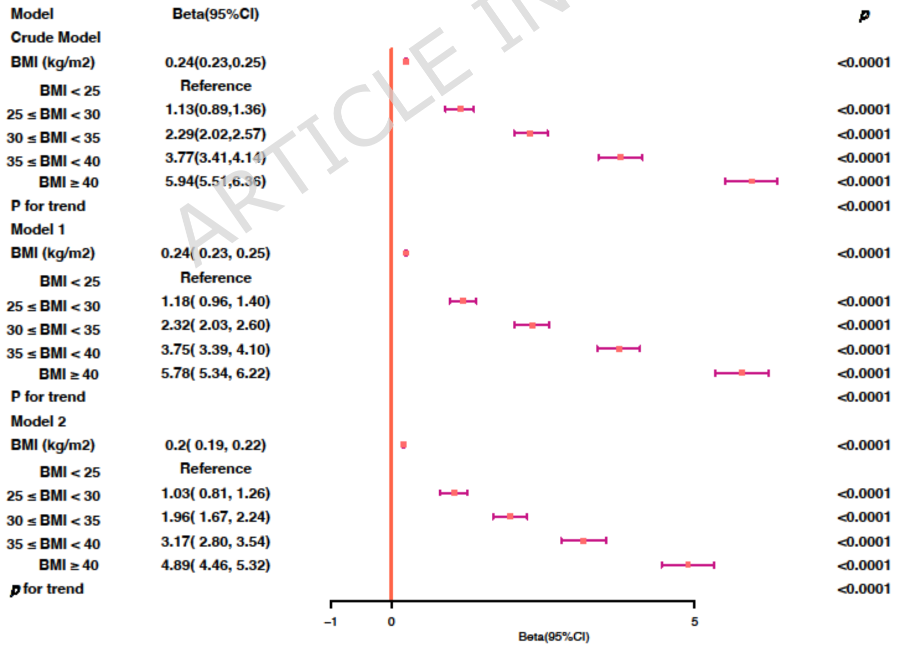


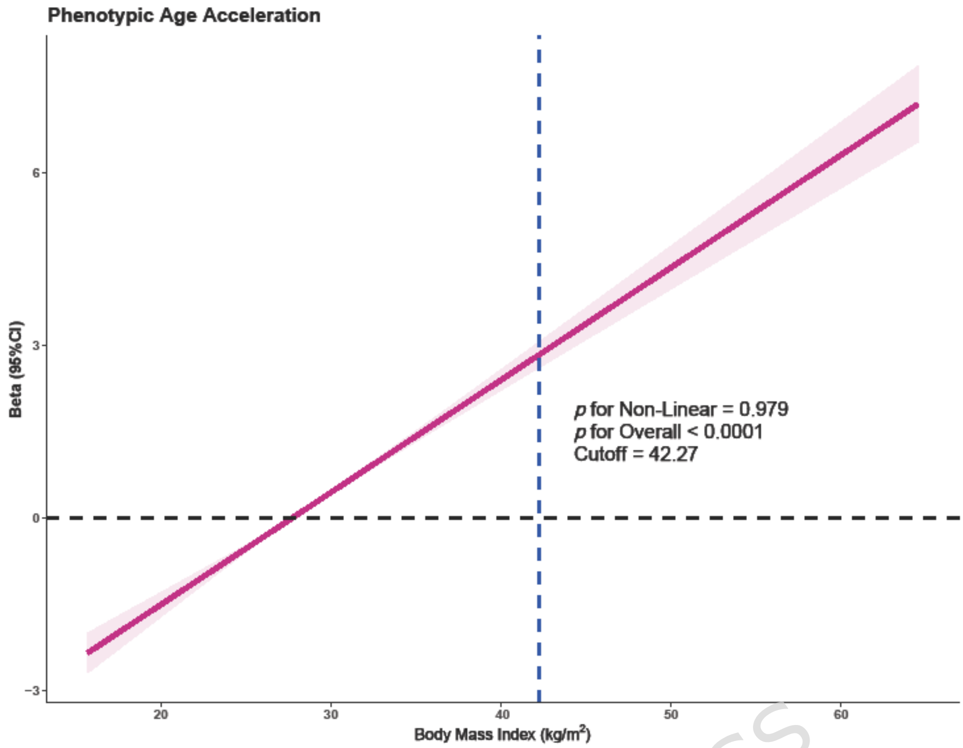
Phenotypic Age Acceleration
7.5
5.0
2.5
0.0
−2.5
Beta (95%CI)
p for Non-Linear = 0.005
p for Overall < 0.0001
Cutoff = 27.37
20
30
40
50
60
Body Mass Index (kg/m2)

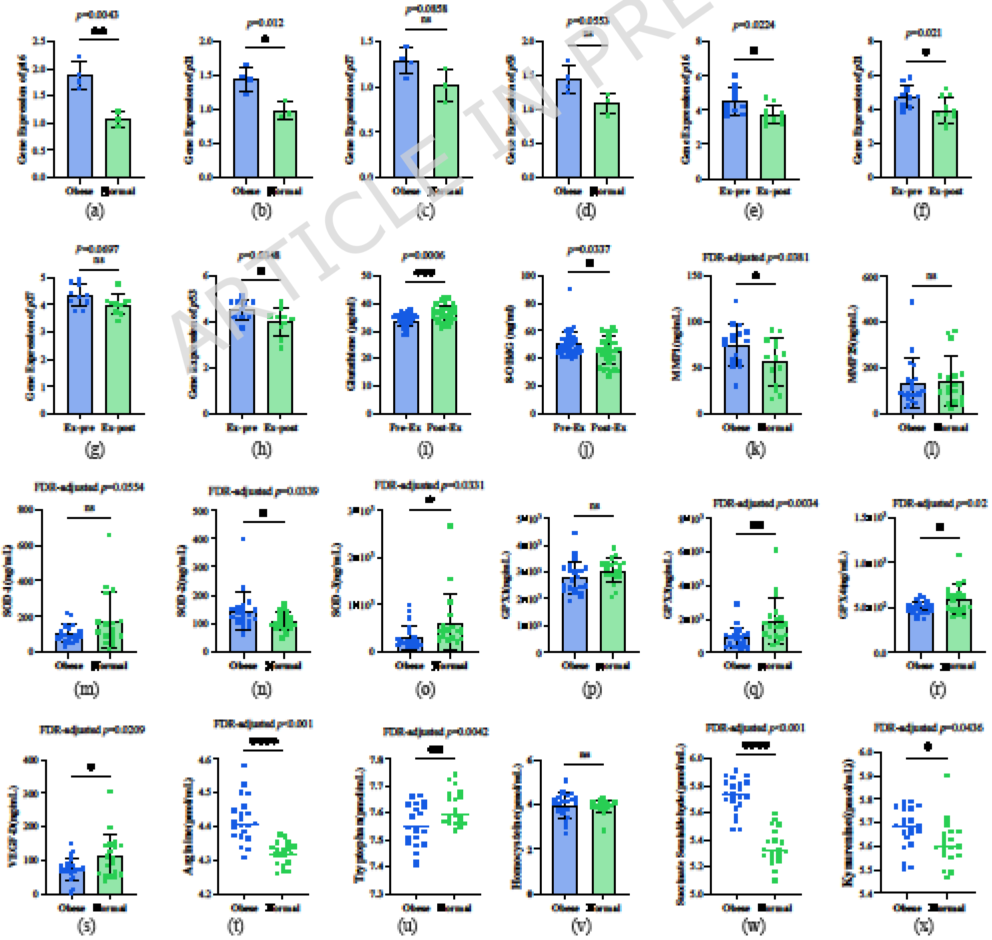


Model Beta(95%CI) p
Crude Model
BMI (kg/m2) 0.21(0.20,0.23) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 0.54(0.30,0.77) <0.0001
30 ≤ BMI < 35 1.72(1.44,2.01) <0.0001
35 ≤ BMI < 40 3.08(2.64,3.51) <0.0001
BMI ≥ 40 5.06(4.52,5.60) <0.0001
P for trend <0.0001
Model 1
BMI (kg/m2) 0.23(0.22, 0.25) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 0.87(0.64, 1.09) <0.0001
30 ≤ BMI < 35 1.99(1.73, 2.24) <0.0001
35 ≤ BMI < 40 3.39(2.96, 3.81) <0.0001
BMI ≥ 40 5.29(4.75, 5.83) <0.0001
P for trend <0.0001
Model 2
BMI (kg/m2) 0.19(0.17, 0.21) <0.0001
BMI < 25 Reference
25 ≤ BMI < 30 0.67(0.44, 0.90) <0.0001
30 ≤ BMI < 35 1.56(1.31, 1.81) <0.0001
35 ≤ BMI < 40 2.71(2.31, 3.10) <0.0001
BMI ≥ 40 4.32(3.79, 4.85) <0.0001
p for trend <0.0001
−1
0
5
Beta(95%CI)


Phenotypic Age Acceleration
Beta (95%CI)
6
3
0
p for Non-Linear = 0.005
p for Overall < 0.0001
Cutoff = 33.4
20
30
40
50
60
Body Mass Index (kg/m²)


Model | Beta(95%CI) | p
Crude Model
BMI (kg/m2) | 0.24(0.23,0.25) | <0.0001
BMI < 25 | Reference
25 ≤ BMI < 30 | 1.13(0.89,1.36) | <0.0001
30 ≤ BMI < 35 | 2.29(2.02,2.57) | <0.0001
35 ≤ BMI < 40 | 3.77(3.41,4.14) | <0.0001
BMI ≥ 40 | 5.94(5.51,6.36) | <0.0001
P for trend | | <0.0001
Model 1
BMI (kg/m2) | 0.24(0.23, 0.25) | <0.0001
BMI < 25 | Reference
25 ≤ BMI < 30 | 1.18(0.96, 1.40) | <0.0001
30 ≤ BMI < 35 | 2.32(2.03, 2.60) | <0.0001
35 ≤ BMI < 40 | 3.75(3.39, 4.10) | <0.0001
BMI ≥ 40 | 5.78(5.34, 6.22) | <0.0001
P for trend | | <0.0001
Model 2
BMI (kg/m2) | 0.2(0.19, 0.22) | <0.0001
BMI < 25 | Reference
25 ≤ BMI < 30 | 1.03(0.81, 1.26) | <0.0001
30 ≤ BMI < 35 | 1.96(1.67, 2.24) | <0.0001
35 ≤ BMI < 40 | 3.17(2.80, 3.54) | <0.0001
BMI ≥ 40 | 4.89(4.46, 5.32) | <0.0001
p for trend | | <0.0001
−1
0
5
Beta(95%CI)


Phenotypic Age Acceleration
Beta (95%CI)
6
3
0
−3
20
30
40
50
60
Body Mass Index (kg/m²)
p for Non-Linear = 0.979
p for Overall < 0.0001
Cutoff = 42.27


(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)
(k)
(l)
(m)
(n)
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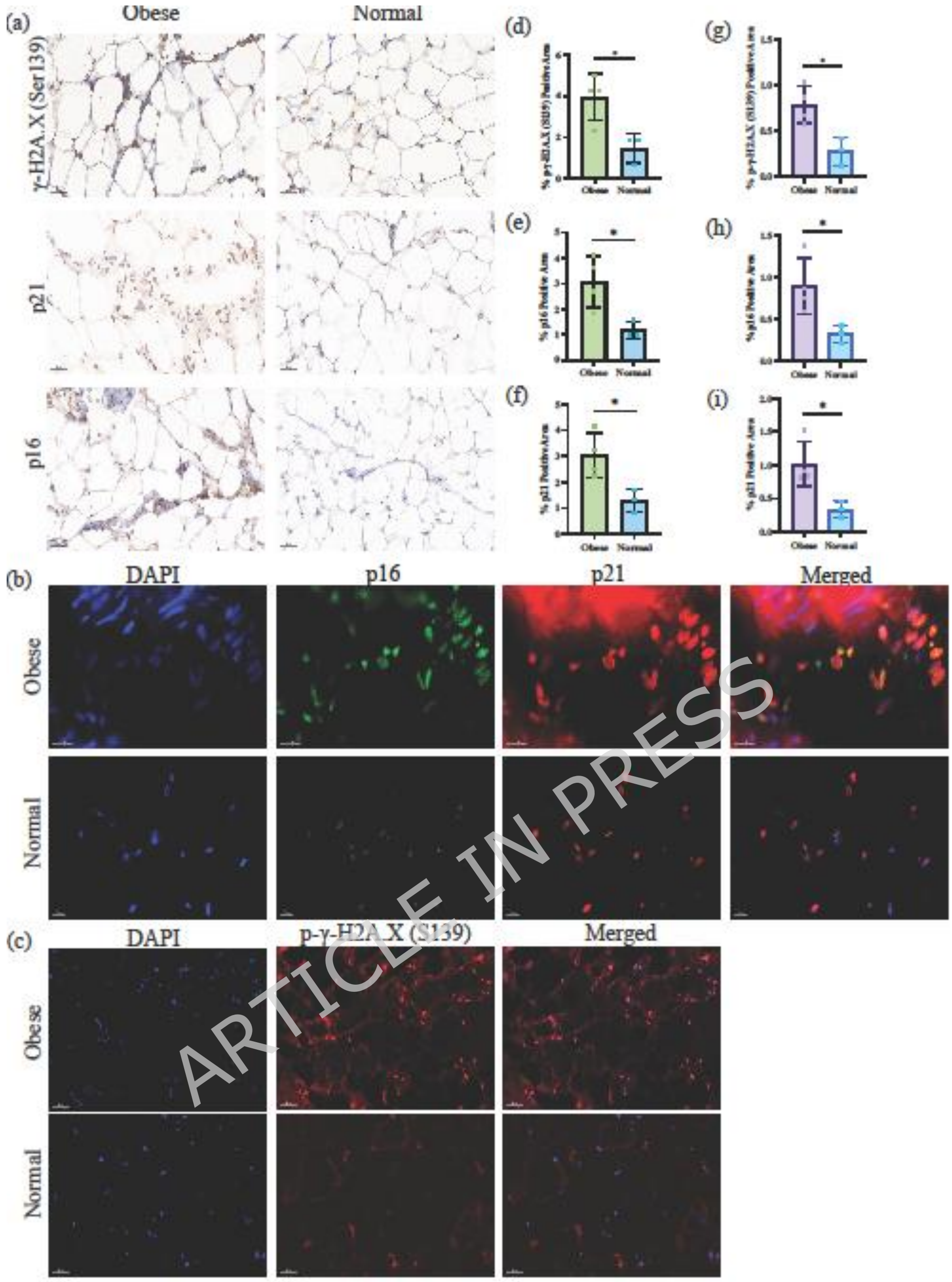
(q)
(r)
(s)
(t)
(u)
(v)
(w)
(x)



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